

Transgenerational effects of obesogenic diets in rodents: A meta-analysis

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Summary

Obesity is a major health condition that affects millions worldwide. There is an increased interest in understanding the adverse outcomes associated with obesogenic diets. A multitude of studies have investigated the transgenerational impacts of maternal and parental obesogenic diets on subsequent generations of offspring, but results have largely been mixed. We conducted a systematic review and meta-analysis on rodent studies to elucidate how obesogenic diets impact the mean and variance of grand-offspring traits. Our study focused on transgenerational effects (i.e., F2 and F3 generations) in one-off and multigenerational exposure studies. From 33 included articles, we obtained 407 effect sizes representing pairwise comparisons of control and treatment grand-offspring groups pertaining to measures of body weight, adiposity, glucose, insulin, leptin, and triglycerides. We found evidence that male and female grand-offspring descended from grandparents exposed to an obesogenic diet displayed phenotypes consistent with metabolic syndrome, especially in cases where the obesogenic diet was continued across generations. Further, we found stronger evidence for the effects of grand-maternal than grand-paternal exposure on grand-offspring traits. A high-fat diet in one-off exposure studies did not seem to impact phenotypic variation, whereas in multigenerational exposure studies it reduced variation in several traits.

KEYWORDS

grand-offspring, grand-parents, obesity, systematic review

1 | INTRODUCTION

Obesity—the excess accumulation of adipose tissue—is a major health condition that leads to impaired physical and psychosocial health and well-being.^{1,2} Obesity affects millions of people

worldwide, and alarmingly, the World Health Organization estimates that global rates of obesity have tripled since 1975³ and continue to rise. Increased food supply and its higher caloric content are major driving factors of this epidemic.⁴ An increasing number of animal studies incorporate obesogenic diets

Abbreviations: F0, parental generation, i.e., first generation exposed to an obesogenic diet (and an appropriate nonexposed control group); F1, first generation offspring, i.e., descendants of F0 animals first exposed to an obesogenic diet (and an appropriate nonexposed control group); F2, second generation of descendants of F0 animals first exposed to an obesogenic diet (and an appropriate nonexposed control group); F3, third generation of descendants of F0 animals first exposed to an obesogenic diet (and an appropriate nonexposed control group); InRR, log-transformed response ratio, i.e., an effect size expressing the ratio of trait means between control and treatment grand-offspring groups; InCVR, log-transformed coefficient of variation ratio, i.e., an effect size expressing the ratio of trait variabilities between control and treatment grand-offspring groups, controlling for a potential mean–variance relationship; PRISMA, Preferred reporting items for systematic reviews and meta-analyses; CI, Confidence interval; PI, Prediction interval.

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to evaluate adverse outcomes of obesity and explore potential remedies.⁵

Obesity has a strong genetic component, but an individual's lifestyle and environment also play a large role in the development of obesity and associated metabolic disorders.^{6,7} However, early life nutrition is thought to be one of the most critical factors.⁸ In mammals, an individual's phenotype can be influenced by the in utero environment, determined by maternal condition and nutrition—a form of “developmental programming.”⁹ Developmental programming describes the phenomenon whereby conditions present during early development render an individual susceptible to metabolic disease later in life.¹⁰ For example, there is accumulating evidence that maternal obesity during pregnancy has adverse effects on offspring health,¹¹ including the increased likelihood of obesity,¹² impaired leptin signalling,¹³ hypertension,¹⁴ hyperglycemia,¹⁵ reduced insulin tolerance,¹⁶ and type 2 diabetes.¹⁷ While the effects of maternal nutrition are well-recognized and supported by extensive empirical research, there is also emerging evidence for strong effects of paternal nutrition.^{18–22} Further, both maternal and paternal effects can be carried beyond the F1 (offspring) generation and this may involve epigenetic mechanisms, which work to alter the phenotype through changes in gene expression rather than eliciting changes to the DNA sequence itself.^{23–27} Epigenetic effects transferred across multiple generations, from the parents (F0) to grand-offspring (F2 and beyond), are termed “transgenerational effects”²⁸ (for “truly transgenerational effects,” see the literature^{29–31}).

There are two main manipulations used in empirical studies of transgenerational effects: one-off exposure and multigenerational exposure (Figure 1A). One-off exposure involves exposing only the F0 generation to the experimental treatment (i.e., obesogenic diet). All generations that follow are devoid of any dietary manipulation (i.e., they are kept on a control diet). In contrast, multigenerational exposure involves exposing not only the F0 animals but also the generations that follow. Such study designs allow researchers to disentangle mechanisms associated with indirect (i.e., epigenetic) and direct (i.e., developmental programming) influences of obesogenic diets. Specifically, one-off exposure experiments expose subtle transgenerational epigenetic changes, while multigenerational exposure experiments show cumulative effects of continuous exposure across generations so that epigenetic influences can interact with direct effects of obesogenic diets on development.

When testing the effects of high-fat diets across generations, one must use an appropriate animal model. Laboratory rodents, such as mice and rats, have been used for over three decades to study various aspects of metabolic syndrome³² because the experimental conditions can be manipulated with ease to observe treatment effects across generations within a relatively short period.³³ Many studies use rodents maintained on a high-fat diet to determine health effects.³⁴ The composition of high-fat diets used can vary widely among studies.³⁵ Reported direct effects on offspring include increased body weight; hyperphagia; adiposity; insulin and leptin resistance^{13,36–38}; impaired glucose tolerance^{39,40}; hypertension⁴¹; and raised plasma

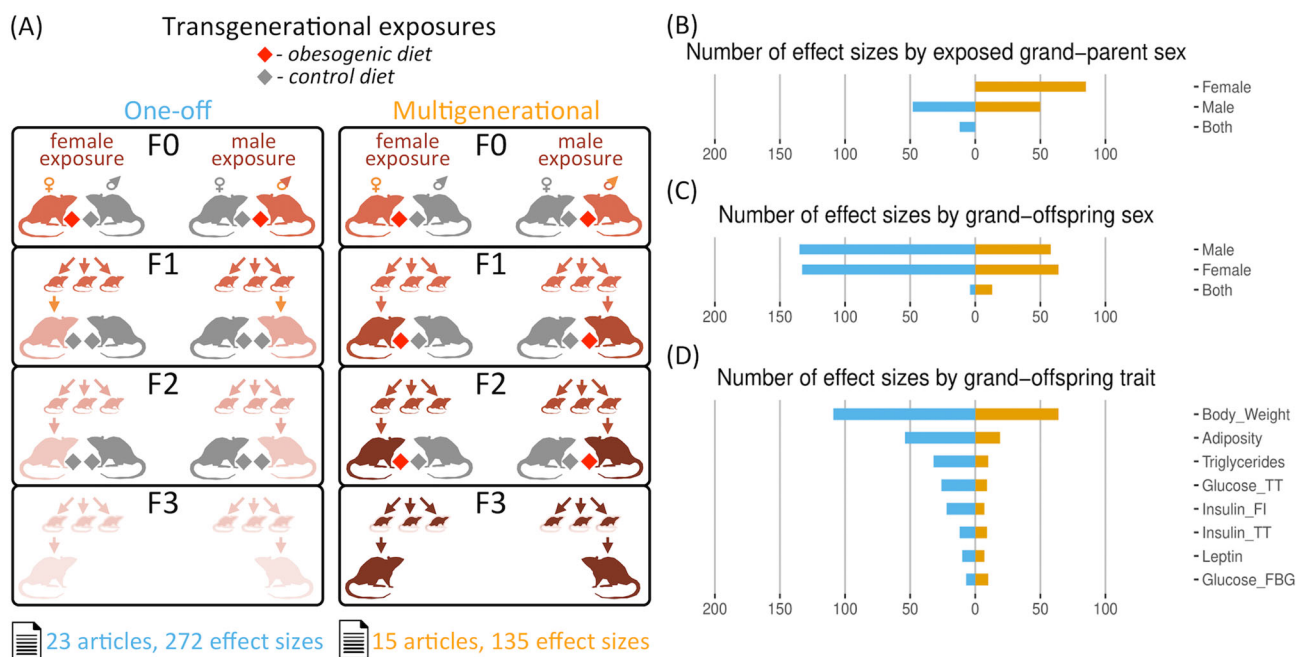


FIGURE 1 (A) Conceptual diagram of two main types of transgenerational experiments included in the meta-analysis. In one-off experiments, only F0 generation adults are exposed to obesogenic diet before and/or during breeding. Then, all animals are kept on control/standard diets. As a result, in subsequent generations the effects of exposure to obesogenic diet is expected to become progressively weaker. In multigenerational experiments, F0 and subsequent generations are exposed to obesogenic diets before and/or during breeding. As result, in subsequent generations, the effects of exposure to obesogenic diets are compounded. (B–D) summaries of the counts of effect sizes for the main analyzed factors in one-off and multigenerational datasets used for meta-analysis

lipids.⁴² Further, there is evidence that these effects can impact offspring in a sex-specific manner.^{43–45}

Despite overwhelming evidence suggesting detrimental impacts associated with obesogenic diets, there is very little work synthesizing studies on transgenerational effects. Such syntheses are critical for understanding the effects of obesogenic diets across generations. Systematic reviews and meta-analyses are standard tools for summarizing empirical studies in many fields. Typically, meta-analytic studies on nutrition and development almost exclusively focus on comparing means between experimental and control groups.⁴⁶ Consequently, inter-subject variability is rarely explored. However, phenotypic variability in obesity has been shown to have an inherited component.^{47,48} Thus, understanding the effects of a given treatment on the variability of an outcome is just as important as understanding its effects on the mean, as variation in biological systems forms the basis of ecological and evolutionary processes (i.e., natural selection). As such, the importance of trait variability has been increasingly argued for in ecology, evolutionary biology,^{46,49} and medical sciences.^{50,51} Recently, a robust method was developed for the analysis of variance in meta-analytic models.⁵² Using this method, as well as an established framework for analyses of the effects on the mean trait values, we conducted a systematic review and meta-analysis of rodent studies (rats and mice strains) in one-off and multigenerational dietary exposure experiments.

Our study aims to address four main questions: (1) What are the overall magnitudes of effects of a one-off grandparental (i.e., F0) and a multigenerational exposure to obesogenic diets on grand-offspring (i.e., F2 and F3)? (2) How do the effects of an obesogenic diet on grand-offspring compare between grand-maternal and grand-paternal exposure? (3) Do the effects of obesogenic diets impact female and male grand-offspring differently? and (4) What traits are most strongly impacted in grand-offspring because of grand-parental obesogenic diet exposure? For each question, we examine the effects on grand-offspring trait mean and variability, comparing control (normal food conditions) and treatment (obesogenic diets) groups.

2 | METHODS

We followed The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines⁵³ in reporting our systematic review and meta-analysis. We provide full details of searches and screening in the Supporting Information. Before searches, we registered a protocol of all our methods on the Open Science Framework (<https://osf.io/sg6wj/register/565fb3678c5e4a66b5582f67>). We present amendments to this protocol, with justifications, in the Supporting Information.

2.1 | Literature search

We performed a comprehensive systematic review of the academic literature using four online databases (Scopus, Medline, and the core

collection in ISI Web of Knowledge, Embase) and other sources (Google Scholar, snowballing). Our initial search took place in April 2018. We also explored grey literature across three online platforms: Trove, OpenGrey, and ProQuest. We performed an update of our searches in May 2020, following the same procedures.

2.2 | Screening and study selection

We screened titles and abstracts of downloaded bibliometric records using Rayyan QCRI.⁵⁴ We aimed to identify experimental studies on wild-type laboratory rodents (excluding strains that were genetically modified or selected for metabolism-related traits), where founders (F0) were exposed to an obesogenic dietary treatment and grand-offspring (F2, F3) phenotypes were reported. Two researchers (HA and AA, and then HA and ML for the search update) independently screened all records to locate potentially relevant studies. We then used the following criteria to screen full-texts of studies that passed the first stage of screening: (1) study had to be empirical work using laboratory rodent models (rats, mice); (2) the rodents used must be from wild-type strains (not mutant, knockdown, or selected for metabolism-related traits, e.g., lean or obese strains); (3) F0-generation rodents of any sex must have been exposed to an obesogenic diet (e.g., high-fat, high-energy, Western diet) alongside a control (not-exposed) group; (4) animals were bred to produce subsequent generations up to a minimum of F2 generation (grand-offspring), and these subsequent generations were kept on a control diet (one-off exposure), or were kept on the diet matching that of their parents (multigenerational exposure); (5) study reported morphological or physiological traits associated with obesity or metabolic syndrome (body weight, adiposity, blood glucose, insulin, leptin, and triglyceride levels) for F2 and/or subsequent generations; (6) study presented relevant data and statistics (mean, standard error/standard deviation, sample sizes) allowing calculation of effect sizes (we attempted contacting authors for missing data and information from recent studies where possible).

2.3 | Data extraction and coding

Our initial extraction and coding captured paper-level information, such as author, title, publication year, and place (see the Supporting Information for the full list of variables and descriptions). We then coded experiment-level information (some studies included more than one experiment or multiple exposure lines), such as rodent species, strain, exposure type (one-off or multigenerational), the composition of obesogenic diets, timing of exposure of F0 generation to obesogenic diet, and sex of exposed animals in the F0 generation. We also coded grand-offspring generation (as F2 or F3), grand-offspring sex, diet, and age at measurement. We collected quantitative data for the traits of interest in the grand-offspring: body weight, adiposity, blood glucose (fasting glucose and glucose tolerance), leptin, and triglyceride levels. For each measurement, we extracted the mean,

standard deviation (or standard error, as reported), and sample size for the treatment group (offspring of grandparents exposed to obesogenic diets) and appropriate control group (Figure 1A). We used the R package *metaDigitise* v.1.0.0⁵⁵ to retrieve quantitative data from figures.

We categorized glucose data into measurements of fasting blood glucose (representing baseline glucose levels), and glucose tolerance tests (representing the organism's response to a sudden increase of glucose in the blood), both used as indicators of diabetes. Tolerance test data were usually presented as tolerance test curves displaying glucose levels over a test period (usually 2 hours), with initial measurements taken as a baseline on fasted animals before glucose injection. We amalgamated tolerance test curve data into a single AUC (i.e., total area under the curve) estimate for each group of animals; we also used initial measurement from each curve as a fasting glucose estimate. Insulin data extractions and processing followed the same procedure, with measurements categorized as plasma/serum fasting insulin or insulin tolerance tests.

We categorized adiposity data according to the body location of adipose tissue (e.g., visceral fat, total fat). Methods to quantify adiposity mainly involved weighing dissected fat pads (10 studies), with some studies employing x-rays (2 studies) and high-resolution imaging (1 study). Values presented as a proportion of body weight were recalculated into grams using associated body weight data (see the Supporting Information). For triglyceride and leptin data, we noted the type of blood extraction (plasma/serum). Additionally, we collected body weight data for F0 founders (where reported) to compare grandparents' body mass around the end of the obesogenic diet treatment to the body mass of their grand-offspring at a similar age. Further details of data extraction and calculations are described in the Supporting Information.

2.4 | Calculating effect sizes

We used the ln-transformed response ratio (*lnRR*)—commonly used in meta-analyses in the biological and medical sciences.^{56,57} It expresses the ratio of trait means between control and treatment grand-offspring groups. To compare variances between control and treatment grand-offspring groups, we used ln-transformed coefficient of variation ratio (*lnCVR*) to control for a potential mean–variance relationship.^{52,58} We calculated these effect sizes and their associated measurement error variances (s^2) following eqs. (5), (6), (11), and (12) in Nakagawa et al.⁵² Positive values of estimated effect sizes can be interpreted as trait mean value or variability being greater in the grand-offspring of parents exposed to obesogenic diets, relative to the grand-offspring of parents not exposed to obesogenic diets.

2.5 | Meta-analysis and meta-regression models

We ran analyses in RStudio v.1.2.1335⁵⁹ using R v.3.6.0.⁶⁰ Statistical models were run using the R package *metafor* v.2.0.⁶¹ We ran

multilevel meta-analytic and meta-regression models, which are extensions of standard random-effects models.⁶² Our dataset had cases where the different sets of treatment groups were compared against the same group of control animals.⁶³ To account for this non-independence, we calculated a variance–covariance matrix (eqs. (19.18) and (19.19) in Olkin and Glesser⁶⁴) and used it as a variance component in meta-analytic and meta-regression models. In all meta-analytic and meta-regression models, we used Paper ID, Rodent Strain ID, Effect Size ID, and Trait as our random factors (except the models where Trait was used as a fixed factor). We calculated overall heterogeneity for meta-analytic models using the multilevel versions of the I^2 statistic.⁶² We conducted all analyses for both *lnRR* and *lnCVR* effect sizes. We performed analyses separately for one-off and multigenerational experiment data because they represent different biological processes and questions. We first estimated overall means for these two datasets using multilevel meta-analytic models (i.e., intercept-only models). We then merged these two datasets to formally compare the magnitudes of estimated average effects of one-off and multigenerational exposures using meta-regression with Exposure Type as a moderator (fixed factor). We ran all subsequent meta-regression analyses separately on one-off and multigenerational datasets. Our main meta-regression models (with one fixed factor at a time) included three moderators: sex of the exposed grandparent (an additional model examined differences between grandparents exposed before mating), sex of grand-offspring and measured trait category. We estimated marginal R^2 values for each meta-regression model following Nakagawa and Schielzeth,⁶⁵ to determine the contribution of these moderators to explaining variation across studies. We created forest-like plots using the *orchaRd* package⁶⁶ to visualise distributions of the raw effect sizes and their mean estimates, together with associated confidence intervals (CI) and prediction intervals (PI). In the text, we converted point estimates into percentage change, for easier interpretation.

2.6 | Full model and model selection analysis

The full meta-regression model included all three fixed effects of interest (with no interactions): sex of exposed grandparents, sex of measured grand-offspring, and measured trait type. Based on this multivariate model, we performed model selection using the *MuMIn* package⁶⁷ to determine the most influential moderator and moderator combinations, for both data sets and both effect size types.

2.7 | Publication bias analyses

We assessed publication bias in the one-off and multigenerational datasets by visually inspecting funnel plots for asymmetry⁶⁸ in the distribution of the residuals of effect sizes (*sensu*⁶²) from a full multivariate model with all three fixed effects. In addition, we performed a modified multilevel-model version of Egger's regression⁶⁹ by including sampling variance in a full meta-regression model. Finally, we

investigated whether there are time trends in the distribution of effect sizes (time lag effect⁷⁰), by using a meta-regression model with publication year as a continuous moderator.

2.8 | Additional analyses

We also ran three additional univariate meta-regression models on both one-off and multigenerational datasets. To explore the effects of the severity of the obesogenic diet on the F0 generation, we analyzed total diet energy (kcal/g), diet protein to non-protein ratio (by weight), and duration of exposure to obesogenic diet (in days). We also tested whether average effects on F2 grand-offspring differ from effects on F3 great grand-offspring. Further, we used body weight data for F0 grandparents around the end of their obesogenic diet treatment (where reported) to compare the effects of obesogenic diets on the body mass of F0 grandparents with the effects in grand-offspring. As a supplement, we also explored the effects of rodent type and period of exposure in females. We conducted all additional analyses for both *lnRR* and *lnCVR* effect sizes (except for F0 grandparent's body weight, in which analyses were only conducted with *lnRR*). All R code and datasets are available at <https://github.com/Apex619/meta-analysis>.

3 | RESULTS

3.1 | Dataset description

Results of our literature search are summarized in the PRISMA diagram in Figures S1–S3 and Tables S1–S3. From the 33 included articles (Table 1), we obtained 407 effect sizes representing pairwise comparisons of control and treatment groups pertaining to body weight, adiposity, glucose, insulin, leptin, and triglycerides in grand-offspring generations F2 and F3. Individual articles contributed between 2 to 57 effect sizes. The measurements were taken from 1,164 and 1,090 unique grand-offspring from treatment and control groups, respectively.

We found 23 articles with one-off exposure type (272 effect sizes) and 15 articles with multigenerational exposure experiments (135 effect sizes; some articles include both types of exposure; see Figure 1). A few articles presented data from the same or very similar experiments, and we categorized these as representing the same study—in total this yielded 19 one-off exposure and 12 multigenerational exposure studies in our data set.

In terms of rodent species used, the dataset comprised of 242 effect sizes from mice (from 18 articles), and 165 from rats (from 15 articles), representing 6 laboratory strains in total. Mice and rats were almost equally represented in one-off and multigenerational exposure studies. The dataset was dominated by experiments where F0 females were exposed (F0 exposed females: 297 effect sizes from 26 studies, F0 exposed males: 98 effect sizes from 7 studies; we had only one study where both females and males were exposed and then bred together to yield F1. Transgenerational data came mainly

from F2 grand-offspring (325 effect sizes from 30 studies), as opposed to F3 great grand-offspring (82 effect sizes from 10 studies). Grand-offspring measurements were distributed evenly between the sexes (females: 197 effect sizes from 23 studies, males: 193 effect sizes from 24 studies, mixed-sex groups: 17 effect sizes from 5 studies).

Body weight was the best-represented offspring trait (173 effect sizes from all 31 studies), followed by adiposity (73 effect sizes from 13 studies). We had fewer data points for triglycerides (42 effect sizes from 14 studies), glucose tolerance tests (35 effect sizes from 13 studies), fasting glucose (17 effect sizes from 8 studies), fasting insulin (29 effect sizes from 10 studies), insulin tolerance test (21 effect sizes from 7 studies), and leptin (17 effect sizes from 8 studies) (it is important to note that studies were mixed regarding the method of measurement, with only a few confirming unstressed measures). Figure 1 presents a summary of numbers of effect sizes in one-off and multigenerational datasets by the exposed grandparent sex (B), grand-offspring sex (C), and grand-offspring trait type (D).

The included studies varied in terms of the type and timing of the obesogenic diet treatments. Energetic value ranged from 4.1 to 5.7 kcal/g (mean 4.9 kcal/g, SD 0.4 kcal/g) in the obesogenic diets and 3.1 to 4.1 kcal/g (mean 3.7 kcal/g, SD 0.3 kcal/g) in the control diets. Protein content ranged from 16 to 30% (mean 23%, SD 3%) by weight in obesogenic diets, and 14 to 27% (mean 20%, SD 3%) in control diets, with protein to nonprotein ratio (by weight) ranging from 0.16 to 0.46 (mean 0.32, SD 0.08) in obesogenic diets, and 0.18 to 0.49 (mean 0.30, SD 0.07) in control diets. These diet parameters were similarly distributed among the data points included in one-off and multigenerational datasets (Figure S4). The duration of grandparental exposure to obesogenic diets ranged from 21 to 140 days (mean 83 days, SD 30 days), for males finishing at mating (day 0) and for females often extending into gestation (31% of studies) and/or even lactation (42% of studies). The distributions of the timing of exposures were generally similar between one-off and multigenerational datasets (Figure S5).

3.2 | Effects of exposure type

Our overall analyses examined the effects of grandparental exposure to an obesogenic diet on grand-offspring separately for one-off and multigenerational exposure data (for both males and females). Although statistically nonsignificant, grand-offspring descended from exposed grandparents in one-off exposure studies tended to have overall mean trait values 9% higher than their control counterparts (*lnRR* = 0.085, CI = −0.076 to 0.247, *p* = 0.301; Figure 2A; Table S4). Grand-offspring descended from exposed grandparents in multigenerational exposure experiments, however, had mean trait values 43% higher than control counterparts. This effect was statistically significant (*lnRR* = 0.358, CI = 0.096 to 0.620, *p* = 0.007; Table S4). The difference between average effects of two exposure types was statistically non-significant (meta-regression model on merged data sets with exposure type as

TABLE 1 Full list of included studies with information on dietary fat (percent of fat by weight), dietary codes and/or dietary sources for both control and obesogenic diets used for FO

Author (Year)	Control diet % fat by weight	Obesogenic diet % fat by weight
Adedeji et al. (2019) ¹⁰⁷	10.3% (not reported)	39.0% (not reported)
Andreas et al. (2019) ¹⁰⁸	5.0% (Rodent Diet 20, PicoLab)	36.0% (58R3, TestDiet)
Armitage et al. (2007) ¹⁰⁹	5.3% (RM3, Special Diet Services)	25.7% (RM3 + lard)
Barbosa et al. (2020) ¹¹⁰	4.0% (AIN-93M)	39.5% (37% lard, Pragsoluções Ltda.)
CastroBarbosa et al. (2016) ¹¹¹	4.2% (Labfor R36, Lactamin)	21.2% (TD.88137/TD.08811, Harlan)
CastroBarbosa et al. (2019) ¹¹²	4.2% (Labfor R36, Lactamin)	21.2% (Harlan)
de Assis et al. (2012) ¹¹³	7.0% (AIN93G)	23.0% (not reported)
Ding et al. (2014) ¹¹⁴	not reported	34.9% (D12492, Research Diets)
Dunn and Bale (2009) ¹¹⁵	5.8% (Purina Lab Diets)	24.0% (Research Diets)
Dunn and Bale (2011) ⁷¹	6.0% (Purina Lab Diets)	24.0% (Research Diets)
Fullston et al. (2012) ¹¹⁶	6.0% (SF04-057, Specialty Feeds)	21.0% (SF00-219, Specialty Feeds)
Giraud et al. (2010) ¹¹⁷	11.0% (Research Diets)	24.0% (D12451, Research Diets)
Gniuli et al. (2008) ¹⁷	3.6% (Allied Feeds)	34.0% (Allied Feeds)
Graus-Nunes et al. (2015) ¹¹⁸	7.0% (PragSoluções)	27.0% (PragSoluções)
Hanafi et al. (2016) ⁷²	4.3% (not reported)	26.5% (not reported)
Harasymowicz et al. (2020) ¹¹⁹	4.3% (D11120103, Research Diets)	34.9% (D11120105, Research Diets)
Hoile et al. (2015) ¹²⁰	5.0% (not reported)	21.0% (not reported)
Huang et al. (2017) ¹²¹	5.0% (not reported)	24.0% (not reported)
King et al. (2013) ¹²²	4.8% (D12328, Research Diets)	35.8% (D12331, Research Diets)
Lannes et al. (2015) ⁷³	7.5% (PragSoluções)	27.0% (PragSoluções)
Li et al. (2012) ¹²³	not reported (Research Diets)	34.9% (Research Diets)
Martins Terra et al. (2019) ¹²⁴	4.0% (CR-1, Nuvelab)	45.1% (PragSoluções)
Masuyama et al. (2015) ¹²⁵	4.2% (Oriental Yeast)	35.0% (Oriental Yeast)
Masuyama et al. (2016) ¹²⁶	4.4% (Oriental Yeast)	35.0% (Oriental Yeast)
Nasu et al. (2007) ¹²⁷	4.6% (Nippon Clea)	32.0% (not reported)
Oshio et al. (2020) ¹²⁸	11.0% (Quintia)	23.9% (PragSoluções)
Park et al. (2018) ¹²⁹	7.0% (Doo Yeol Biotech)	27.0% (Doo Yeol Biotech)
Sarker et al. (2018) ¹³⁰	4.5% (3430, Kliba-Nafag)	35.0% (2127, Kliba-Nafag)
Schellong et al. (2020) ¹³¹	3.3% (V1534-000, ssniff R/M-H)	15.5% (132006, Altromin)
Tait et al. (2015) ¹³²	5.0% (58B0 (AIN-76A) Purina)	24.0% (58V8 (D12451), Purina)
Thompson et al. (2019) ¹³³	5.0% (20, Pico Laboratory Rodent)	36.0% (58R3, Test Diet)
Winther et al. (2019) ¹³⁴	4.2% (D12450 J, Research Diets)	34.9% (D12492, Research Diets)
Zhang et al. (2019) ¹³⁵	7.0% (not reported)	20.7% (not reported)

moderator: $\ln RR$ difference = -0.187 , $CI = -0.256$ to -0.118 ; Table S5). Total heterogeneity among effect sizes was high for both one-off and multigenerational datasets ($I^2 = 95.7\%$ and 99.2% , respectively; Table S4), warranting analyses of moderators to explain the variation in the effect sizes for effects on mean trait values.

The average effects on trait variability for both types of exposure were small and statistically non-significant (one-off data: increase of 3%, $\ln CVR = 0.033$, $CI = -0.134$ to 0.200 , $p = 0.698$; multigenerational data: decrease of 7%, $\ln CVR = -0.074$, $CI = -0.282$ to 0.134 , $p = 0.486$; Figure 2B; Table S4; $\ln CVR$

difference = -0.024 , $CI = -0.173$ to 0.124 , $p = 0.750$, $R^2 = 0.001$; Table S5). Total data heterogeneity was also moderately high for both one-off and multigenerational data ($I^2 = 63.0\%$ and 71.6% , respectively; Table S4).

3.3 | Effects of exposed grandparents sex (FO)

In one-off exposure experiments, we found no clear effect of the sex of exposed grandparents on mean trait values of grand-offspring ($\ln RR$ difference = -0.017 , $CI = -0.065$ to 0.032 , $p = 0.508$, $R^2 = 0.001$;

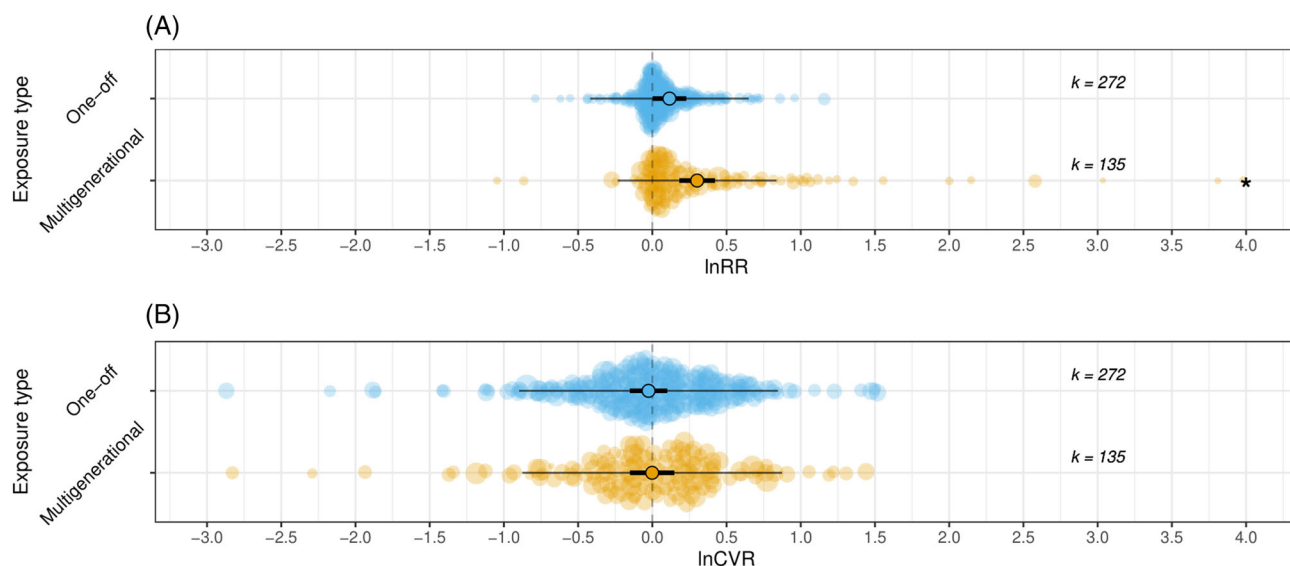


FIGURE 2 Forest-like (orchard) plots of effect size estimates from meta-regression model with experiment type (one-off or multigenerational) as a moderator: (A) effects on the mean values of grand-offspring traits ($\ln RR$), and (B) effects on the variances of grand-offspring traits ($\ln CVR$). Thick horizontal lines indicate 95% confidence intervals (CI), thin horizontal lines indicate 95% prediction intervals (PI), with mean estimates at the center; k are the numbers of effect sizes. Pale blue and orange circles represent individual effect sizes, with circle sizes scaled accordingly to precision (weights). Statistically significant mean effect sizes (CI not crossing zero) are marked with an asterisk

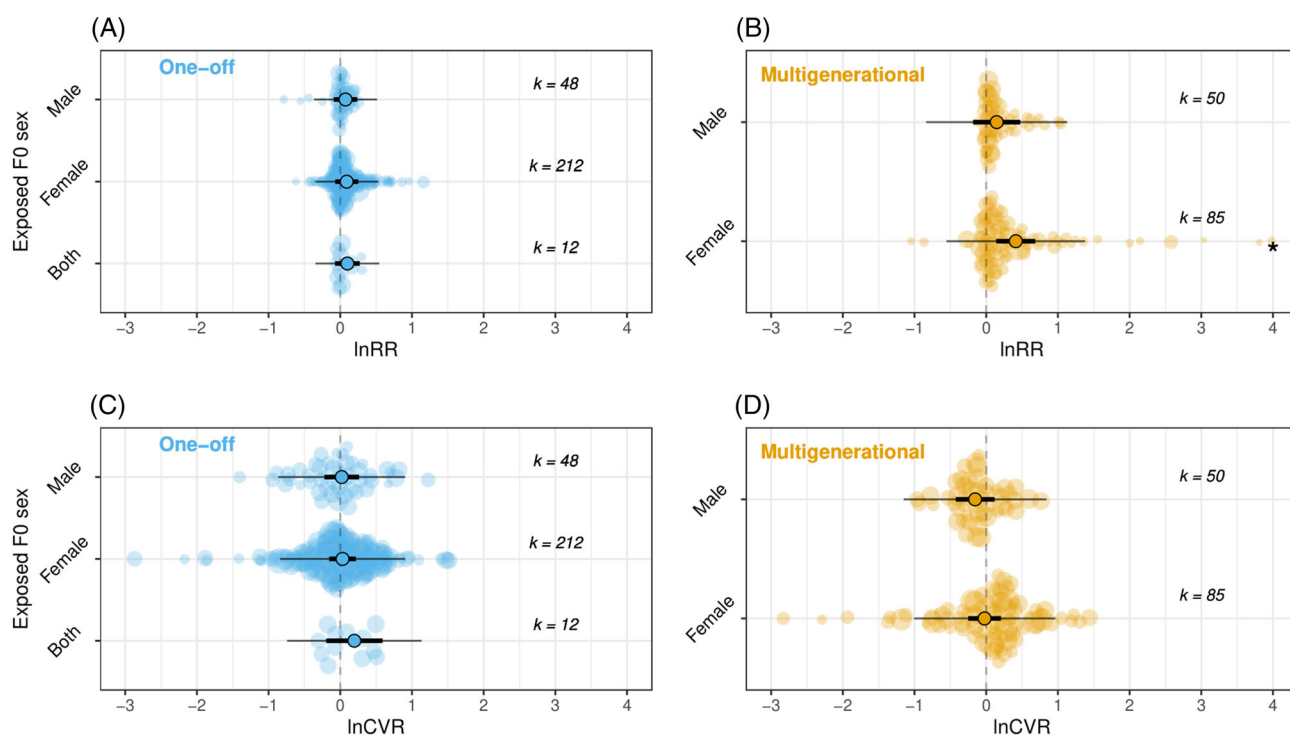


FIGURE 3 Forest-like (orchard) plots of effect size estimates from meta-regression models with the sex of exposed grandparents as a moderator, for one-off and multigenerational datasets: (A,B) effects on the mean values of grand-offspring traits ($\ln RR$), and (C,D) effects on the variances of grand-offspring traits ($\ln CVR$). Thick horizontal lines indicate 95% confidence intervals (CI), thin horizontal lines indicate 95% prediction intervals (PI), with mean estimates at the center; k are the numbers of effect sizes. Pale blue and orange circles represent individual effect sizes, with circle sizes scaled accordingly to precision (weights). Statistically significant mean effect sizes (CI not crossing zero) are marked with an asterisk

Table S6) and the meta-regression model explained less than 1% of the variation among effect sizes (Figure 3A; Table S6). In addition, there was also no clear effect of the sex of grandparents exposed only before mating (Table S20). Similarly, effects on variability of grand-offspring traits ($\ln\text{CVR}$) were indistinguishable between grand-maternal and grand-paternal one-off exposure lines ($R^2 = 0.006$; Figure 3C; Table S6). This was also the case for grandparents exposed only before mating (Table S20).

In multigenerational exposure experiments, grand-maternal exposure to an obesogenic diet had a large and statistically significant effect on mean trait values of grand-offspring (grandmothers: increase of 51%, $\ln\text{RR} = 0.412$, $\text{CI} = 0.140$ to 0.683 , $p = 0.003$; Figure 3B; Table S6). The same was true for grand-maternal exposure before mating (Table S20). In contrast, grand-paternal exposed lines were associated with a small and statistically non-significant effect on mean trait values of grand-offspring (grandfathers: increase of 16%, $\ln\text{RR} = 0.146$, $\text{CI} = -0.183$ to 0.475 , $p = 0.384$; Figure 3B) compared to more marked effects from maternal exposure lines ($\ln\text{RR}_{\text{difference}} = -0.266$, $\text{CI} = -0.517$ to -0.015 , $p = 0.038$). This meta-regression model explained 7% of variation among effect sizes. Grand-paternal exposure before mating, however, was associated with a statistically significant effect on grand-offspring, albeit smaller than grand-maternal exposure before mating (Table S20). The effects on variability of grand-offspring traits were indistinguishable from

zero for both grand-maternal and grand-paternal exposures, with no statistically significant difference between them ($\ln\text{RR}_{\text{difference}} = -0.132$, $\text{CI} = -0.420$ to 0.157 ; $R^2 = 0.017$; Figure 3D; Table S6).

3.4 | Grand offspring sex effects

For one-off exposures, the effects on both granddaughter and grandson mean trait values were small and statistically not different from zero, or each other (granddaughters: 10%, $\ln\text{RR} = 0.091$, $\text{CI} = -0.056$ to 0.238 , $p = 0.226$; grandsons: 11%, $\ln\text{RR} = 0.103$, $\text{CI} = -0.044$ to 0.250 , $p = 0.170$; Figure 4A; Table S7). The meta-regression model explained only 0.3% of variation among effect sizes. There was also no effect of grand-offspring sex on trait variability (Figure 4C; Table S7).

For multigenerational exposures, the effects on granddaughter and grandson mean trait values were large and statistically different from zero (granddaughters: 45%, $\ln\text{RR} = 0.373$, $\text{CI} = 0.087$ to 0.658 , $p = 0.010$; grandsons: 39%, $\ln\text{RR} = 0.331$, $\text{CI} = 0.044$ to 0.618 , $p = 0.024$; Figure 4B; Table S7). On average, there was no statistically significant difference between effects on granddaughters and grandsons ($\ln\text{RR}_{\text{difference}} = -0.042$, $\text{CI} = -0.157$ to 0.074 , $R^2 = 0.002$). There was also no apparent difference between granddaughters and

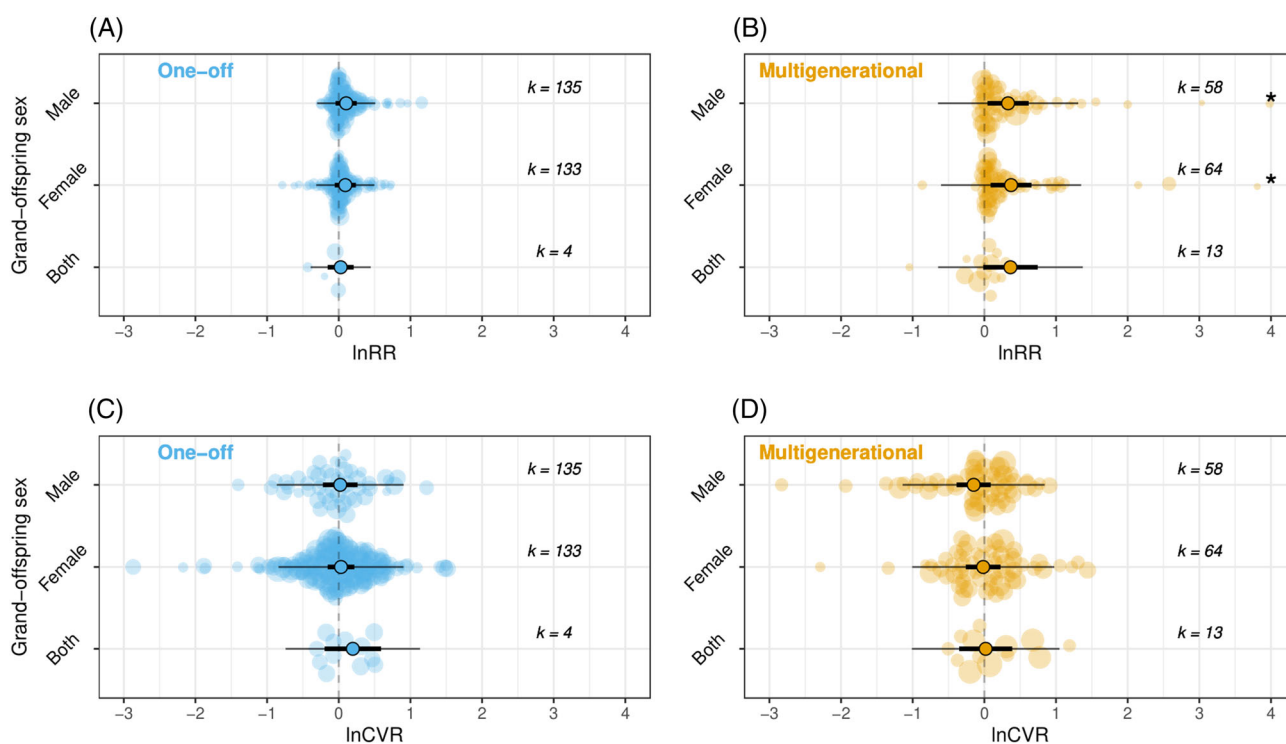


FIGURE 4 Forest-like (orchard) plots of effect size estimates from meta-regression models with the sex of measured grand-offspring as a moderator, for one-off and multigenerational datasets: (A,B) effects on the mean values of grand-offspring traits ($\ln\text{RR}$), and (C,D) effects on the variances of grand-offspring traits ($\ln\text{CVR}$). Thick horizontal lines indicate 95% confidence intervals (CI), thin horizontal lines indicate 95% prediction intervals (PI), with mean estimates at the center; k are the numbers of effect sizes. Pale blue and orange circles represent individual effect sizes, with circle sizes scaled accordingly to precision (weights). Statistically significant mean effect sizes (CI not crossing zero) are marked with an asterisk

grandsons in the average effect on variability of the traits ($\ln RR$ difference = -0.133 , CI = -0.381 to 0.116 ; $R^2 = 0.002$; Figure 4D; Table S7).

3.5 | Effects of offspring trait category

For one-off exposures, grand-offspring of grandparents exposed to obesogenic diets were on average more obese than offspring from unexposed lines, although the effect was small (20%, $\ln RR = 0.182$, CI = 0.025 to 0.338 , $p = 0.023$; $R^2 = 0.093$; Figure 5A; Table S8). While also small, the average concentration of leptin was increased in grand-offspring of grandparents fed obesogenic diets (20%, $\ln RR = 0.205$, CI = 0.029 to 0.381 , $p = 0.022$). Average effects on adiposity and leptin levels were significantly larger than in the

remaining trait categories (Table S8). The effects on trait variability ($\ln CVR$) were usually small and not statistically different from zero for all trait categories ($R^2 = 0.064$; Figure 5C, Table S8).

Similar to what was observed in one-off exposures, after multigenerational exposure to obesogenic diets, grand-offspring had significantly, and remarkably higher levels of adiposity (121%, $\ln RR = 0.794$, CI = 0.578 to 1.010 , $p < 0.001$; Figure 5B; Table S8). They also had higher levels of fasting insulin (58%, $\ln RR = 0.457$, CI = 0.138 to 0.776 , $p = 0.005$), leptin (139%, $\ln RR = 0.872$, CI = 0.577 to 1.168 , $p < 0.001$) and triglycerides (41%, $\ln RR = 0.340$, CI = 0.088 to 0.592 , $p = 0.008$). The average effects on adiposity and leptin levels were significantly larger than in the remaining traits ($R^2 = 0.370$; Table S8). Surprisingly, the effect on average body mass was small and not statistically significant (11%, $\ln RR = 0.100$, CI = -0.062 to 0.262 , $p = 0.225$). The effects of multigenerational

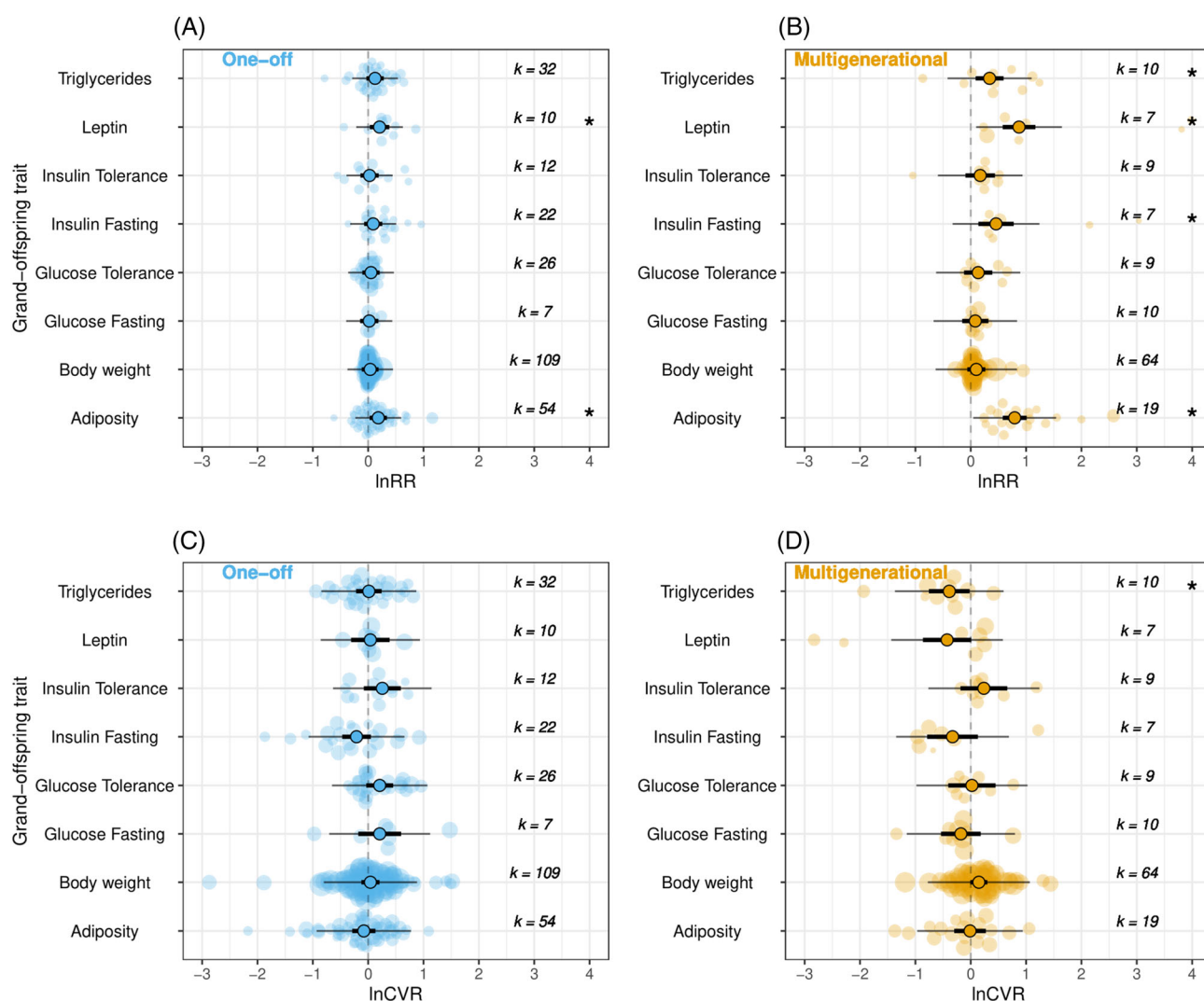


FIGURE 5 Forest-like (orchard) plots of effect size estimates from meta-regression models with the grand-offspring trait type as a moderator, for one-off and multigenerational datasets: (A,B) effects on the mean values of grand-offspring traits ($\ln RR$) and (C,D) effects on the variances of grand-offspring traits ($\ln CVR$). Thick horizontal lines indicate 95% confidence intervals (CI), thin horizontal lines indicate 95% prediction intervals (PI), with mean estimates at the center; k are the numbers of effect sizes. Pale blue and orange circles represent individual effect sizes, with circle sizes scaled accordingly to their precision (weights). Statistically significant mean effect sizes (CI not crossing zero) are marked with an asterisk

exposure on grand-offspring trait variability were small and statistically non-significant for all traits except one (Figure 5D, Table S8). Namely, grand-offspring of parents exposed to obesogenic diets across at least two generations had less variability in levels of blood triglycerides than their nonexposed counterparts ($\ln\text{CVR} = -0.388$, $\text{CI} = -0.756$ to -0.020 , $p = 0.039$; $R^2 = 0.166$).

3.6 | Full model and model selection analysis

The full meta-regression model included sex of exposed grandparents, sex of measured grand-offspring, and measured trait type (Table S9). Although there was still a great deal of model uncertainty, these three moderators jointly explained 13% of the variation in the effects on mean trait values in one-off exposure data, 38% in multigenerational data, and 8% of the effects on trait variability, in both datasets. Model selection analysis indicated that trait type was the most influential moderator of average effect sizes on the trait means in one-off data, and for both trait means and variabilities in multigenerational data (Table S10).

3.7 | Publication bias analyses

Visual inspection of enhanced-contour funnel plots of residuals did not show clear data distribution skewness indicative of publication bias (Figure S6). A variant of Egger's regression test using full multi-level meta-regression models indicated significant funnel plot asymmetry only for the multigenerational $\ln\text{RR}$ dataset (Table S11). Finally, the slope of the linear regression between publication year and the effect size was not significantly different from zero for all data sets (Table S12).

3.8 | Additional analyses

We performed three additional analyses: (1) to examine the effects of diet composition and exposure time; (2) compare average effects between F2 and F3 offspring; and (3) compare F0 and grand-offspring body mass. Results of additional models examining effects of rodent type and period of exposure in females can be found in Tables S19 and S21, respectively.

Diets with higher total energy content in multigenerational experiments had larger effects on mean grand-offspring trait values ($\ln\text{RR}_{\text{slope}} = 0.188$, $\text{CI} = 0.062$ to 0.313 , $p = 0.003$; $R^2 = 0.153$; Table S13). Neither relative protein content of the obesogenic diet, nor the duration of the exposure of grandparents to the obesogenic diet, appeared to significantly influence the magnitude of effect sizes in one-off and multigenerational datasets and effect measures (Tables S14 and S15).

Moreover, we found no difference between the average magnitudes of the effects between grand-offspring from F2 and F3 generations (Table S16).

Exposed grandparents from both one-off and multigenerational datasets were, on average, 15% heavier than their non-exposed counterparts ($\ln\text{RR} = 0.139$, $\text{CI} = 0.062$ to 0.216 ; Tables S17 and S18). For

comparison, predicted differences between their grand-offspring at similar age (around 100 days old) would be around 7% in one-off experiments, and around 17% after multigenerational experiments (but note very wide confidence and prediction intervals; Tables S17 and S18).

4 | DISCUSSION

We addressed four main questions relating to the effects of obesogenic diets on morphological and physiological traits of grand-offspring from F2 and F3 generations. First, we have shown that grand-offspring in multigenerational exposure lines had mean trait values 43% larger than control grand-offspring. We have also shown an analogous, although weaker, trend in one-off exposure experiments. Second, we show that grand-maternal effects were larger than grand-paternal effects in multigenerational experiments, but not in one-off exposure experiments (9% larger). Third, the effects on grandsons and granddaughters were similar in both multigenerational and one-off exposure experiments. Fourth, we found that adiposity was the most affected grand-offspring trait in both types of exposure experiments. In multigenerational experiments, leptin, fasting insulin, and triglyceride levels were also elevated by the obesogenic diet treatment, with weaker effects on body weight. Also, effects on grand-offspring trait variability were usually small and statistically nonsignificant, apart from triglyceride levels in multigenerational experiments, where inter-subject variation was lower in treatment grand-offspring in comparison to control counterparts. Notably, leptin, fasting insulin, and fasting glucose exhibited a similar trend that inter-subject variability appears lower in the treatment group. We discuss these findings, and additional insights, in detail below.

4.1 | One-off vs. multigenerational exposure

We found no significant effects on grand-offspring traits of treatment animals from one-off experiments, although the effect was in the expected direction. Others have shown that subtle effects of obesogenic diets can persist beyond the F0 generation even without further diet manipulation,⁷¹⁻⁷⁴ whereas our meta-analysis highlights that there is uncertainty and heterogeneity regarding this effect. In contrast, grand-offspring traits of treatment animals from multigenerational experiments were 43% higher than their control counterparts. This result matched our expectations that effects would be exacerbated when obesogenic diets are applied across generations, because observed changes in grand-offspring traits result from the cumulative influence (or interaction) of trans-generational, inter-generational, and direct nutritional effects. Notably, three-quarters of the animals in multigenerational studies were fed a high-fat diet around the time of trait measurements. As such, direct effects of grand-offspring diet could explain, in part, the large differences in trait means between experimental and control grand-offspring, while potentially masking more subtle purely trans-generational effects.

4.2 | Grand-parental (F0) sex effects

For one-off exposure experiments, impacts on grand offspring traits did not depend on the sex of exposed grandparents (i.e., there was no effect of F0 sex, regardless of exposure period). The lack of a statistically significant grandparental sex effect in one-off exposure experiments is in line with the fact that effect sizes are both small and highly heterogeneous, and there are relatively few studies with grand-paternal exposures (5 out of 18 articles with one-off exposure). This lack of bias towards either sex has been shown before in a diet-induced obesity study (intergenerational) which noted equal strength of epigenetic inheritance from both male and female gametes.⁷⁵ Certainly, the same may be happening on a transgenerational scale, especially if offspring in question are not exposed to an obesogenic diet. In contrast, for multigenerational experiments, grand-offspring from exposed grandmothers had mean trait values 23% higher than treatment grand-offspring from exposed grandfathers. This result reflects more limited opportunities for transmitting nutritional insults by males at each generation. In mammals, mothers can easily pass on nutritional insults to their offspring during gestation and lactation. Fathers can influence their offspring only via sperm and seminal fluid during mating. Because of the latter, paternal one-off and transgenerational studies would hint more clearly towards epigenetic mechanisms, which modify the epigenome of sperm.⁷⁶

4.3 | Grand-offspring (F2 and F3) sex effects

The average magnitudes of the dietary exposure effects were similar between female and male grand-offspring for both one-off and multigenerational exposure studies. This is surprising because previous studies examining the transgenerational effects of obesogenic diets have shown sex-specific effects.^{27,77,78} Although mechanisms underlying sex-specific effects remain poorly understood, it has been suggested that one sex may be more sensitive than the other due to the role of sex chromosomes⁷⁹ and factors acting during development, such as ontogenetic changes in gene expression.^{80,81} Our work shows that, on average, there is no consistent pattern of sex-dependent offspring vulnerability to obesogenic diets, at least in the assessed offspring traits and dietary conditions. Further work is needed to clarify this given that sex-dependent responses can depend on the type of diet.^{82,83}

4.4 | Grand-offspring (F2 and F3) trait type effects

Our analyses yielded three interesting findings for the effects in different categories of grand-offspring traits. Firstly, we found adiposity to be the most affected trait. We expected that treatment offspring would display abnormally high levels of adiposity in response to developmental programming by obesogenic diets, especially in multigenerational experiments. In line with this prediction, treatment grand-offspring had on average 20% more fat than control offspring in one-off exposure experiments and 121% more fat in multigenerational experiments. Unexpectedly, the increase in body fat

was not accompanied by an equivalent increase in body weight. Treatment grand-offspring tended to be only 4% heavier than control grand-offspring in one-off experiments, and 11% heavier in multigenerational experiments. This discrepancy can be partially attributed to the fact that only 3 studies, out of 12 studies with adiposity data, reported total fat measurements using imaging techniques. The remaining studies reported adipose mass measurements for fat pads, which could be differentially affected by nutrition.

Secondly, physiological traits that are functionally linked to adipose tissue showed similar patterns of effects as adiposity measurements. As adipose tissue stores triglycerides,⁸⁴ it is unsurprising that we found a parallel increase of triglyceride levels in treatment grand-offspring in multigenerational experiments. Treatment grand-offspring in multigenerational experiments also had significantly greater levels of leptin (139% higher). Leptin is one of the major players involved in energy homeostasis. It is produced by adipocytes and has a central role in the regulation of food intake and body weight.⁸⁵ However, individuals with obesity not only have elevated leptin levels but also develop “leptin resistance,” a complex phenomenon, where increased circulating leptin does not reduce appetite and body mass.^{86,87} We showed that treatment grand-offspring in multigenerational experiments also had significantly greater levels of fasting insulin (58% higher). Increased insulin is required to maintain normal glucose tolerance, also known as compensatory hyperinsulinemia.⁸⁸ This potentially explains why we did not observe abnormal glucose levels in multigenerational exposure experiments. In one-off exposure experiments, the overall pattern of effects was similar, but all effects were small and statistically non-significant. Finally, taken together, the above results suggest weak transgenerational inheritance of the metabolic syndrome, a cluster of conditions that increase the risk of heart disease and diabetes.⁸⁹ Large effects observed in multigenerational exposure experiments, likely arise predominantly due to maternal effects and direct offspring diet effects. The surprisingly small effect on body weights coupled with large effects on physiology in multigenerational experiments is in line with the concept of “normal weight obesity,” which highlights the need to stratify risk based on underlying metabolic factors, such as adiposity, and metabolic changes, rather than body weight alone.⁹⁰ Moreover, since many of the included papers reported body weights of grandparents, we were able to show that when controlling for age, the magnitude of the effect of obesogenic diet in multigenerational exposure experiments was similar for grandparents and grand-offspring (15% and 17%, respectively). This result suggests a lack of strong cumulative effects of multigenerational exposure, at least on body weight.

Thirdly, our study also examined differences in variation between control and treatment grand-offspring. Effects on grand-offspring trait variability were small and not statistically significant, apart from triglyceride levels in multigenerational experiments. While work in flies showed that high-fat diets increase phenotypic variation,⁹¹ we found that triglyceride levels varied significantly less in treatment than in the control grand-offspring (32% difference). While non-significant, it is noteworthy that leptin, fasting insulin, and fasting glucose also generally followed this pattern. One

possible explanation is that the effects of a high-fat diet are impacted by a ceiling effect whereby levels are elevated to their physiological capacity, effectively lowering variation and masking potential differences between individuals. Such a scenario was previously proposed to explain the lack of increase in glucose concentrations of rodents under a high-fat diet.⁹²

4.5 | Additional findings

We investigated whether obesogenic diet composition and duration of grand-parental exposure to such a diet can moderate the effects on grand-offspring traits in both one-off and multigenerational datasets. The minimum amount of protein (by weight) to support adequate growth and reproduction is ~16% in mice and ~14% in rats.⁹³ Studies included in our meta-analysis had adequate, but not particularly high protein content, with protein by weight % ranging from 16% to 30%. We found no effect of protein content in the obesogenic diets on grand-offspring trait values. This finding is surprising given the expectation that low protein intake may decrease thermogenesis and satiety, as well as lead to an increase in subsequent energy intake.⁹⁴ However, our result is consistent with a study showing that protein content is not a major factor in regulating energy intake and causing an obesogenic response (i.e., adiposity), rather dietary fat is.⁹⁵ In line with this, we revealed that experimental diets with higher total energy content (usually containing more fat) had a significantly larger impact on mean trait values of grand-offspring in multigenerational experiments. Foods high in energy have been shown to have a robust and significant effect on satiety and satiation, thereby facilitating overconsumption of fat, leading to obese phenotypes.⁹⁶ It has also been established in previous studies that longer exposure to obesogenic diet results in larger effects on morphological and physiological traits in directly exposed animals.^{97–100} However, we found no statistically significant influence of duration of grand-parental diet exposure on the magnitude of effect sizes in grand-offspring, indicating that this effect may get diluted as it is passed to subsequent generations. We also found no difference between F2- and F3-generation grand-offspring in the average magnitude of effects (for both one-off and multigenerational studies), indicating weak, but durable, transgenerational transfer of dietary effects.

4.6 | Limitations and future directions

Our meta-analysis has several limitations. Firstly, we worked with unevenly distributed numbers of available effect sizes when attempting to answer our questions regarding exposure type (66% of the data for one-off exposure studies), F0 sex effects (74% of the data from grand-maternal exposure), and grand-offspring trait effects (besides body weight, all other traits did not exceed 17% of effect sizes). All studies (except one) reported body weights, but not necessarily other traits, depending on the study's focus. Further, most studies did not report outcomes pertaining to the F0 generation itself (i.e., adiposity). This made it difficult to answer other questions

concerning the effects of grandparental exposure to an obesogenic diet (i.e., separating the effects of dietary exposures and the resulting changes in the bodies of parents). In addition, a lack of effect sizes from specific exposure windows (i.e., pregnancy and lactation) also prevented us from answering more questions pertaining to the influences of an obesogenic diet (i.e., on body weight). For example, it has been shown in that variation in litter size during the lactation period plays a key role in determining weaning weight and postweaning growth rate in mice¹⁰¹; as well as body weight and adiposity in rats.¹⁰² Future studies can benefit by including experimental groups which disentangle these exposure windows.

Secondly, given the subtlety of effects in certain cases (i.e., one-off exposure studies had an average effect of 9%), large sample sizes are required to detect changes due to experimental manipulations in empirical studies. In our data sets, numbers of measured grand-offspring in treatment groups ranged from 3 to 34, with a mean of 13 and a median of 10, thus limiting power to detect small effects. The magnitudes of effect sizes from our meta-analysis can be used in power calculations to guide the appropriate sample sizes required in future empirical studies.¹⁰³

Thirdly, truly transgenerational effects are those that are found in generations that were not exposed to the factor that triggered the change in phenotype.²⁹ In other words, the effects can be considered truly transgenerational only if grand-offspring have no direct contact with the grand-parental environment. In mammals, this definition is important when comparing grand-maternal and grand-paternal effects. When an F0 mother is exposed to an obesogenic diet, the developing offspring (F1) is directly affected in utero and during lactation, as well as F2 germ cells inside these developing F1 offspring.³⁰ As such, the F2 generation may already be at an adverse risk of developing metabolic disease, and a minimum F3 generation is required to detect true transgenerational effects. When an F0 father is exposed, their germline (future F1) is exposed, but not F2 germ cells. Thus, in grand-paternal exposure studies, F2 is sufficient for detecting true transgenerational effects.^{29,31} In our dataset, approximately 80% of effect sizes were from F2 grand-offspring. The remaining 20% of effect sizes is from F3 grand-offspring: 82 effect sizes overall, including 58 from one-off exposure experiments (only 44 from great-grand-maternal exposure). Given that our dataset is dominated by grand-maternal exposures and F2 grand-offspring data, it is hard to completely disentangle truly transgenerational mechanisms from developmental programming effects. However, our analyses also show that the average magnitude of observed effects did not significantly differ between F2 and F3 grand-offspring, indicating that the difference between transgenerational effects in F2 and “truly transgenerational effects” in F3 is also subtle.

Fourthly, the pre-planned moderators we used in our meta-regression analyses did not effectively account for the high level of variation present in effect sizes across studies, with R^2 values ranging from 0 to 0.37. Improved reporting, as well as a consistent framework for study designs, may help limit heterogeneity, and facilitate more detailed analyses. As such, we highly recommend following updated guidelines for reporting animal research.¹⁰⁴

Lastly, our publication bias tests indicated some potential funnel plot asymmetry, but only in the multigenerational *lnRR* (mean trait values) dataset. Funnel plot asymmetry may stem from true unexplained biological heterogeneity, not publication bias.⁶⁹ In any case, quality reporting including all results, regardless of their outcome, is recommended for the primary studies and care should be taken when interpreting meta-analytical results.¹⁰⁵

To our knowledge, our meta-analysis is the first to examine differences in variance between obesogenic and control group diets. As most meta-analyses focus on mean differences, we hope to open another avenue of meta-analytical exploration, as the difference in variance are not only medically relevant,¹⁰⁶ but also evolutionary and ecologically relevant,⁵² as natural selection processes act on variation.

5 | CONCLUSIONS

This meta-analysis on rodent studies aimed to address how an obesogenic diet impacts the mean and variance of metabolic traits across generations. Overall, we have demonstrated that the grand-offspring of exposed grandparents display phenotypes consistent with the metabolic syndrome, especially if the effect has been exacerbated by multigenerational exposure to obesogenic diets. Maternal factors have the strongest influence via developmental programming, and both male and female offspring are equally susceptible. Furthermore, the caloric density of the diet plays a significant role in promoting an obesogenic phenotype, and certain metabolic traits, such as adiposity, are more reliable as indicators of metabolic syndrome transfer across generations. A high-fat diet may also result in a ceiling effect, reducing the amount of phenotypic variation in grand-offspring in multigenerational exposure groups. Further empirical and meta-analytical research is needed to elucidate mechanisms underlying true transgenerational inheritance of effects of obesogenic diets, particularly for maternal exposure, as most included studies fail to extend beyond the F2 generation.

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CONFLICT OF INTEREST

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SUPPORTING INFORMATION

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